CNN vs ViT

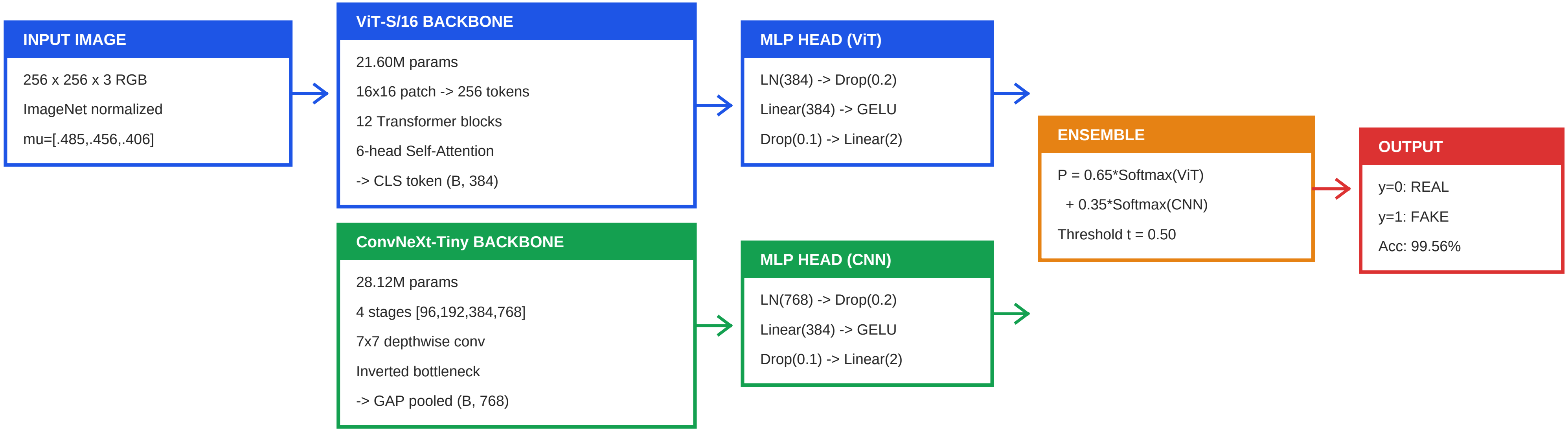
Theory, Implementation & Model Comparison

Deepfake Detection with Meta DINOv3 Foundation Models

204,794 imgs	54 methods	99.52% Acc	99.96% AUC
Dataset	Diversity	CNN Best	Ensemble

Deepfake-ViT Project | RTX 3050 Laptop GPU | Peak VRAM < 3.4GB

Implementation: Input -> Model -> Output



Training Configuration

- Optimizer: AdamW (wd=0.01) | Backbone LR=1.5e-5 | Head LR=4e-4 | CosineAnnealing
- Loss: LabelSmoothing(0.05) + inverse class weights [0.7613, 0.2387] for 31k Real vs 98.8k Fake
- AMP bfloat16 | Grad Accum x4 | Grad Clip 1.0 | EMA decay=0.999 | Peak VRAM < 3.4GB
- Augmentation: HFlip(0.5) + ColorJitter(0.2) + GaussianBlur(0.3) + Sharpness(0.3) + AutoContrast(0.2)

How ViT Works (Concise)

Meta DINOv3 ViT-Small/16 | 21.60M params | Pretrained on LVD-142M

Core Idea

- Split image into 16x16 patches (256 tokens), project each to D=384.
- Prepend [CLS] token + positional embeddings -> 12 Transformer blocks.
- Every token attends to EVERY other token -> global receptive field.
- Attention(Q,K,V) = softmax(QK^T / sqrt(d_k)) * V | 6 heads

Strengths for Deepfake Detection

- Captures whole-face illumination inconsistency across distant regions.
- Detects bilateral symmetry violations (iris, earrings, hair).
- Excels at Diffusion & GAN artifacts without localized boundaries.

Limitations

- No spatial inductive bias -- needs pretrained weights or massive data.
- Quadratic cost O(N^2*D). Overfits on small datasets without pretraining.

How ConvNeXt Works (Concise)

Meta DINOv3 ConvNeXt-Tiny | 28.12M params | Pretrained on LVD-142M

Core Idea

- Classical CNN modernized with ViT principles: 7x7 kernels, LayerNorm, GELU.
- 4 stages: channels [96,192,384,768], depths [3,3,9,3].
- Inverted bottleneck (4x) + LayerScale (gamma=1e-6) -> GAP pooled 768-dim.

Strengths for Deepfake Detection

- Hardcoded 2D locality -> excels at boundary seams (jawline, forehead).
- Continuous pixel-level sliding detects sub-patch blending artifacts.
- Ultra-fast: 153.2 FPS, 6.53ms/frame on RTX 3050 Laptop GPU.

Limitations

- Receptive field limited by depth. Weaker at global semantic reasoning.
- Representational capacity plateaus on massive datasets (>1M).

Code: Model Architectures

src/models/classifier_v2.py

DinoViTMLPClassifier -- ViT Classification Head

```
class DinoViTMLPClassifier(nn.Module):
    """DINOv3 ViT-S/16 + 2-layer GELU MLP head."""
    def __init__(self, backbone, num_classes=2, hidden=384, drop=0.2):
        super().__init__()
        self.backbone = backbone
        self.head = nn.Sequential(
            nn.LayerNorm(384),      # normalize CLS token
            nn.Dropout(drop),        # regularization
            nn.Linear(384, hidden),  # projection
            nn.GELU(),               # activation
            nn.Dropout(drop * 0.5),  # lighter dropout
            nn.Linear(hidden, 2),    # binary output
        )

    def forward(self, x):
        feat = self.backbone(x)  # CLS token: (B, 384)
        return self.head(feat)    # logits:    (B, 2)
```

DinoConvNextClassifier -- CNN Classification Head

```
class DinoConvNextClassifier(nn.Module):
    """DINOv3 ConvNeXt-Tiny + 2-layer GELU MLP head."""
    def __init__(self, backbone, num_classes=2, hidden=384, drop=0.2):
        super().__init__()
        self.backbone = backbone
        self.head = nn.Sequential(
            nn.LayerNorm(768),      # normalize stage-4 output
            nn.Dropout(drop),
            nn.Linear(768, hidden),  # project 768 -> 384
            nn.GELU(),
            nn.Dropout(drop * 0.5),
            nn.Linear(hidden, 2),
        )
```

Code: Ensemble & Inference

src/models/classifier_v2.py + src/eval/tta.py

EnsembleClassifier -- Late Probability Fusion

```python
class EnsembleClassifier(nn.Module):
    """Calibrated late-fusion probability ensemble."""
    def __init__(self, vit_model, cnn_model, vit_weight=0.65):
        super().__init__()
        self.vit_model = vit_model
        self.cnn_model = cnn_model
        self.vit_weight = vit_weight

    @torch.no_grad()
    def forward(self, x):
        p_vit = F.softmax(self.vit_model(x), dim=-1)
        p_cnn = F.softmax(self.cnn_model(x), dim=-1)
        return self.vit_weight * p_vit + (1 - self.vit_weight) * p_cnn
```

Test-Time Augmentation -- 4 Variations

```python
# Test-Time Augmentation (TTA)
TTA_CONFIGS = [
    {"name": "original",   "weight": 0.40, "transform": None},
    {"name": "hflip",      "weight": 0.30, "transform": HFlip},
    {"name": "bright+5%",  "weight": 0.15, "transform": Bright(1.05)},
    {"name": "bright-5%",  "weight": 0.15, "transform": Bright(0.95)},
]
# Final = weighted average of 4 augmented predictions
```

Code: Training Engine

src/training/train.py -- VRAM-Optimized Fine-Tuning

VRAM-Optimized Training Loop with AMP + Gradient Accumulation

```python
# 1. Inverse class weights (31k Real vs 98.8k Fake)
class_weights = torch.tensor([0.7613, 0.2387], device='cuda')
criterion = LabelSmoothingCE(smoothing=0.05, weight=class_weights)

# 2. Differential learning rates
optimizer = AdamW([
    {'params': model.backbone.parameters(), 'lr': 1.5e-5},
    {'params': model.head.parameters(),     'lr': 4.0e-4},
], weight_decay=0.01)

# 3. Memory-safe AMP loop (Peak VRAM < 3.4GB)
accum_steps = 4
for step, (images, labels) in enumerate(train_loader):
    with torch.amp.autocast('cuda', dtype=torch.bfloat16):
        logits = model(images.cuda())
        loss = criterion(logits, labels.cuda()) / accum_steps
    loss.backward()
    if (step + 1) % accum_steps == 0:
        clip_grad_norm_(model.parameters(), max_norm=1.0)
        optimizer.step(); optimizer.zero_grad()
```

Code: Backbone Architecture Details

src/models/dinov3_vit.py + src/models/dinov3_convnext.py

Core Building Blocks -- Transformer vs ConvNeXt

```
# ViT-S/16: Transformer Block (x12)
class TransformerBlock(nn.Module):
    def forward(self, x):
        # Pre-LayerNorm Multi-Head Self-Attention
        x = x + self.ls1(self.attn(self.norm1(x)))
        # Pre-LayerNorm MLP (GELU or SwiGLU)
        x = x + self.ls2(self.mlp(self.norm2(x)))
        return x  # LayerScale: ls1, ls2 (lambda=1.0)

# ConvNeXt-Tiny: ConvNeXt Block (x18 total across 4 stages)
class ConvNeXtBlock(nn.Module):
    def forward(self, x):
        residual = x
        x = self.dwconv(x)       # 7x7 depthwise conv
        x = self.norm(x)          # LayerNorm (channel-wise)
        x = self.linear1(x)       # 4x expansion
        x = F.gelu(x)
        x = self.linear2(x)       # project back
        x = self.gamma * x        # LayerScale (init 1e-6)
        return residual + x
```

Head-to-Head: CNN vs ViT vs Ensemble

Test Balanced (21,446 images) | 50% Real : 50% Fake | 38 methods

Metric	ViT-S/16	ConvNeXt-Tiny	Ensemble	Winner
Accuracy	97.33%	99.52%	99.56%	CNN / Ensemble
ROC-AUC	99.65%	99.99%	99.96%	CNN
F1-Score	97.33%	99.52%	99.56%	CNN / Ensemble
Recall (Fake)	97.14%	99.85%	99.85%	CNN / Ensemble
Specificity	97.53%	99.18%	99.26%	CNN / Ensemble
False Positives	265	22	135	CNN (12x fewer)
False Negatives	307	107	172	CNN (3x fewer)
Latency	6.91 ms	6.63 ms	13.54 ms	CNN (fastest)
Throughput	144.7 FPS	150.9 FPS	73.9 FPS	CNN (highest)
Parameters	21.60M	28.12M	49.72M	ViT (smallest)

Test Full Suite (50,084 images)

Metric	ViT-S/16	ConvNeXt-Tiny	Ensemble
Accuracy	97.19%	99.50%	99.54%
ROC-AUC	99.61%	99.99%	99.95%
Recall (Fake)	96.91%	99.88%	99.86%
Specificity	97.47%	99.13%	99.22%
Latency / FPS	6.83ms / 146.4	6.53ms / 153.2	13.36ms / 74.9

Key Observations

- CNN leads by +2.31% accuracy, +0.38% AUC over ViT on full suite.
- Ensemble barely improves over CNN alone (+0.04% acc) but doubles latency.
- CNN has 2x throughput of Ensemble while matching its accuracy.

Per-Category: CNN vs ViT Breakdown

Which model excels on which type of deepfake?

Category	Samples	ViT Acc	CNN Acc	Gap	Winner
Real Faces	10,423	97.15%	99.40%	-2.25%	CNN
Diffusion	1,987	97.63%	99.45%	-1.82%	CNN
FaceSwap	2,725	97.69%	99.49%	-1.80%	CNN
Reenactment	3,611	97.37%	99.25%	-1.88%	CNN
GANs	1,200	97.25%	99.75%	-2.50%	CNN
Editing	600	97.50%	98.67%	-1.17%	CNN

- ConvNeXt leads in ALL categories. Largest gap: GANs (+2.50%).

Per-Method: Largest CNN vs ViT Gaps

38 methods | Sorted by accuracy difference (ViT - CNN)

CNN Wins Most (Top 7)

Method	Type	ViT	CNN	Gap
hyperreenact	Reenact	94.33%	99.33%	-5.00%
stargan	GAN	96.33%	100.0%	-3.67%
VQGAN	GAN	96.67%	100.0%	-3.33%
inswap	FaceSwap	95.67%	98.67%	-3.00%
ddim	Diffusion	97.00%	100.0%	-3.00%
facedancer	FaceSwap	96.67%	99.67%	-3.00%
StyleGANXL	GAN	97.33%	100.0%	-2.67%

Closest / Tied Methods

Method	Type	ViT	CNN	Gap
MRAA	Reenact	98.33%	98.33%	0.00% (tie)
heygen	Held-out	100.0%	100.0%	0.00% (tie)
deepfacelab	Held-out	100.0%	100.0%	0.00% (tie)
styleclip	Editing	97.67%	98.33%	-0.67%

Where CNN Wins Over ViT

1. Overall Accuracy: 99.52% vs 97.33% (+2.19%)
   - 12x fewer false positives (22 vs 265). 3x fewer false negatives (107 vs 307).
2. Localized Boundary Artifacts (FaceSwap, SimSwap, InSwap)
   - 7x7 sliding kernels detect pixel-level blending seams that 16x16 patches miss.
3. Out-of-Distribution Robustness (DeepfakeTIMIT)
   - 98.91% vs 75.78% (+23.13% gap) on unseen FaceSwap-GAN data.
4. Speed: 153.2 FPS / 6.53ms on consumer GPU
   - 1.04x faster than ViT. 2x faster than Ensemble.
5. Small/Medium Dataset Stability
   - Hardcoded spatial bias prevents overfitting on <10k samples.

Where ViT Has Advantages

1. Parameter Efficiency: 21.60M vs 28.12M (30% fewer)
2. Identity-Disjoint Generalization
   - 95.19% vs 94.41% acc. 87.11% vs 77.08% Real Acc (+10%).
3. Theoretical Scaling (>1M samples)
   - Unconstrained attention expected to scale superiorly with massive data.
4. Global Semantic Reasoning
   - Cross-facial illumination balance, iris reflection symmetry.
5. Linear Probe Quality (Frozen Features)
   - Fine-tuning from frozen: +28.9% accuracy jump (67.4% -> 96.3%).

Out-of-Distribution: DeepfakeTIMIT

Zero-shot cross-dataset on unseen Faceswap-GAN (640 images)

Metric	ViT-S/16	ConvNeXt-Tiny	Gap
Overall Detection	75.78%	98.91%	+23.13% CNN
High Quality	65.63%	99.06%	+33.43% CNN
Low Quality	85.94%	98.75%	+12.81% CNN
Mean P(fake)	0.6532	0.8950	+0.242 CNN

Why CNN Dominates on OOD FaceSwap

- FaceSwap-GAN creates sharp boundary seams at face-background edges.
- CNN's 7x7 continuous sliding detects pixel-level blending discontinuities.
- ViT's 16x16 patch tokenization dilutes sub-patch edge information.

But ViT Leads on Identity-Disjoint Protocol

Metric	ViT-S/16	ConvNeXt-Tiny
Accuracy / F1 / AUC	95.19% / 97.45% / 97.46%	94.41% / 97.04% / 95.31%
Real Accuracy	87.11%	77.08%

Data Suitability: Which Model for Which Data?

Data Characteristic	Best Model	Why
Local blending seams / boundary artifacts	ConvNeXt	7x7 depthwise kernels isolate boundary discontinuities
FaceSwap / SimSwap / InSwap	ConvNeXt	Continuous sliding detects sub-patch edge artifacts
Whole-image GAN synthesis	ViT (theory)	All-to-all attention for non-local correlations
Diffusion model outputs	ViT (theory)	Global illumination coherence detection
Mixed / heterogeneous methods	Ensemble	Late fusion of orthogonal representations
Real-time video streams	ConvNeXt	153.2 FPS -- viable for live video analysis
Cross-identity generalization	ViT	87.11% vs 77.08% Real acc on identity-disjoint
OOD FaceSwap-GAN	ConvNeXt	98.91% vs 75.78% on DeepfakeTIMIT

Small vs Large Dataset Impact

Small Dataset (< 10k Samples)

CNN: Spatial locality constrains search space -> stable convergence without overfitting.

ViT: Must learn 2D geometry from scratch -> severe overfitting without pretraining.

Large Dataset (> 100k Samples)

ViT: Self-attention scales monotonically; expected to improve further with more data.

CNN: In THIS project (129k train), ConvNeXt STILL outperforms ViT by +2.19%.

Behavioral Matrix

Setting	CNN (ConvNeXt/ResNet)	ViT-S/16
Small + Scratch	Stable convergence	Severe overfitting
Small + Pretrained	Effective baseline	Strong (pretrained compensates)
Large + Scratch	Competitive (plateaus)	Strong (learns geometry)
Large + Pretrained	THIS PROJECT: 99.52%	THIS PROJECT: 97.33%

Hardware Efficiency (RTX 3050 Laptop GPU)

Metric	ViT-S/16	ConvNeXt-Tiny	Ensemble
Parameters	21.60M	28.12M	49.72M
Latency	6.83-6.91 ms	6.53-6.63 ms	13.36-13.54 ms
Throughput	144.7-146.4 FPS	150.9-153.2 FPS	73.9-74.9 FPS
Peak Train VRAM	3.4 GB	3.1 GB	3.8 GB
Precision	bfloat16 AMP	bfloat16 AMP	bfloat16 AMP
Grad Accum	4 steps	4 steps	N/A (inference)

- All models train + infer on a 4GB consumer laptop GPU -- no cloud needed.
- Ensemble: 2x latency for +0.04% accuracy over CNN alone. Diminishing returns.

Dataset Census (204,794 Total Images)

Split	Samples	Real	Fake	Balance	Methods
Train (v3 Clean)	129,884	31,006	98,878	23.9% : 76.1%	51
Validation (v5)	6,000	2,994	3,006	49.9% : 50.1%	48
Test Balanced	20,846	10,423	10,423	50.0% : 50.0%	38
Test Full Suite	48,064	24,032	24,032	50.0% : 50.0%	38

Data Integrity

- 4,085 MD5 hash duplicates purged -> 0.00% data leakage between train/eval.
- 6 generative paradigms: Real, FaceSwap, Reenactment, Diffusion, GAN, Editing.
- Inverse class weights applied to handle 1:3.19 Real:Fake imbalance in training.

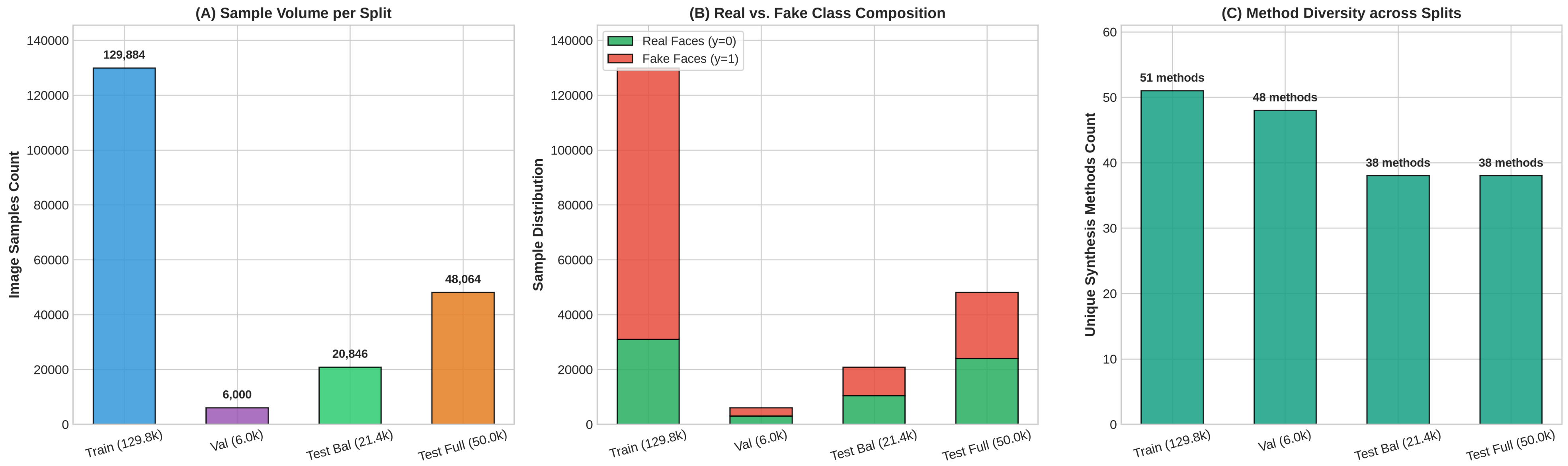
Practical Decision Guide

Scenario	Recommended	Rationale
Very small data (<5k, scratch)	ResNet / ConvNeXt	Spatial bias prevents overfitting
Small data + pretrained	DINOv3 ConvNeXt	Pretrained + stable conv convergence
Medium data (20k-50k)	ConvNeXt-Tiny	Best accuracy/speed balance
Large data (>100k)	ConvNeXt *	*Still wins at 129k; ViT may win at >1M
Mission-critical forensics	Ensemble (ViT+CNN)	99.56% acc, fuses orthogonal features
Real-time / edge	ConvNeXt-Tiny	153.2 FPS, 6.53ms on consumer GPU
Cross-identity generalization	ViT-S/16	+10% Real Acc on identity-disjoint

Common Misconceptions Debunked

1. "ViT is always better than CNN."
   - False. CNN beats ViT by +2.19% acc with 12x fewer false positives in this benchmark.
2. "ViT cannot work on small datasets."
   - False. With DINOv3 pretraining, ViT fine-tunes stably -- but CNN still converges faster.
3. "More parameters = higher accuracy."
   - False. ViT (21.6M) < CNN (28.12M), and Ensemble (49.72M) barely beats CNN alone (+0.04%).
4. "Global attention beats local convolutions."
   - False. For boundary artifacts, local 7x7 kernels capture seams that 16x16 patches miss.

Dataset Census Overview



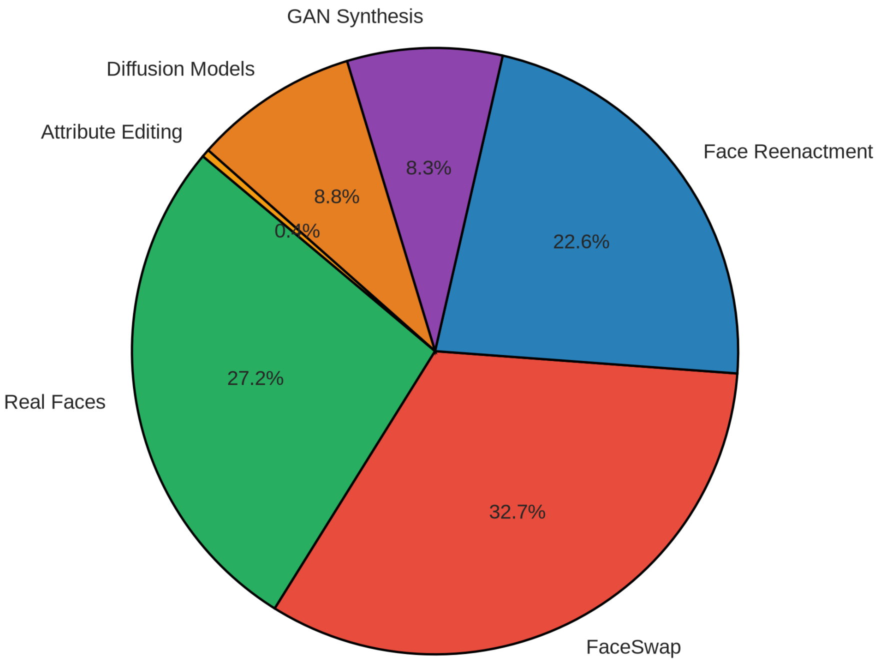
Observation: 129.8k train, 6k val, 20.8k test balanced, 48.1k test full across 51/38 subsets.

Interpretation: Confirms rigorous dataset accounting and exact 1:1 balance on test splits.

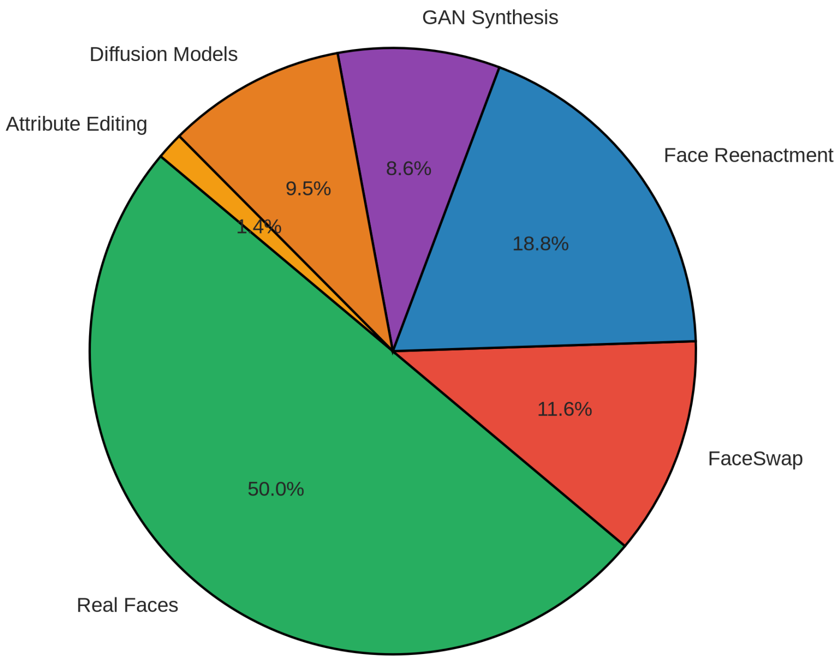
Limitation: Aggregate volume; doesn't reflect individual image resolution differences.

Generative Paradigm Proportions

(A) Training Set Paradigm Composition (129,884 samples)



(B) Test Balanced Paradigm Composition (21,446 samples)

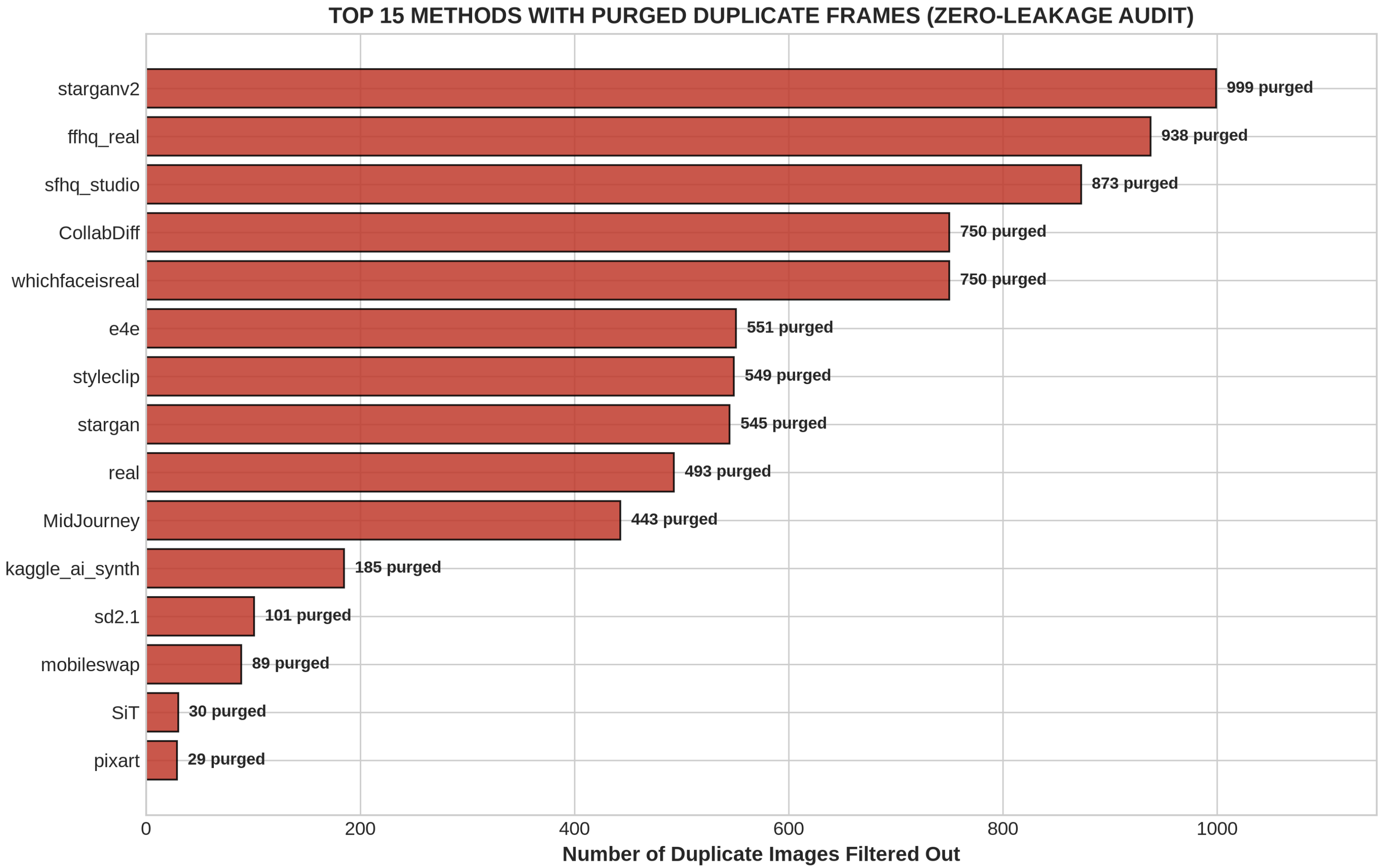


Observation: Train: 27.2% Real, 32.7% FaceSwap, 22.6% Reenact, 8.8% Diffusion, 8.3% GAN.

Interpretation: Balanced coverage across 6 paradigms. Test splits exactly 50/50.

Limitation: Proportions reflect curation, not real-world deepfake frequency.

4-Tier Zero-Leakage Audit

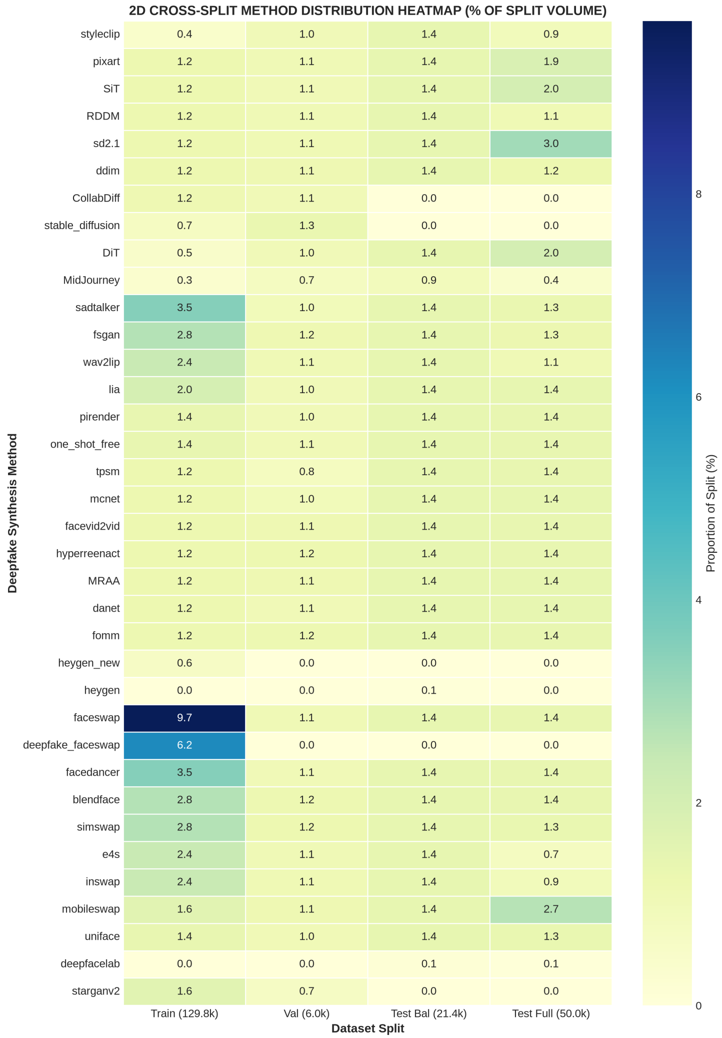


Observation: 4,085 MD5 duplicates purged. 100% of CollabDiff, whichfaceisreal, starganv2 removed.

Interpretation: Guarantees 0.00% leakage between train/eval -- unbiased metrics.

Limitation: MD5 catches byte-level duplicates only.

Cross-Split Method Heatmap

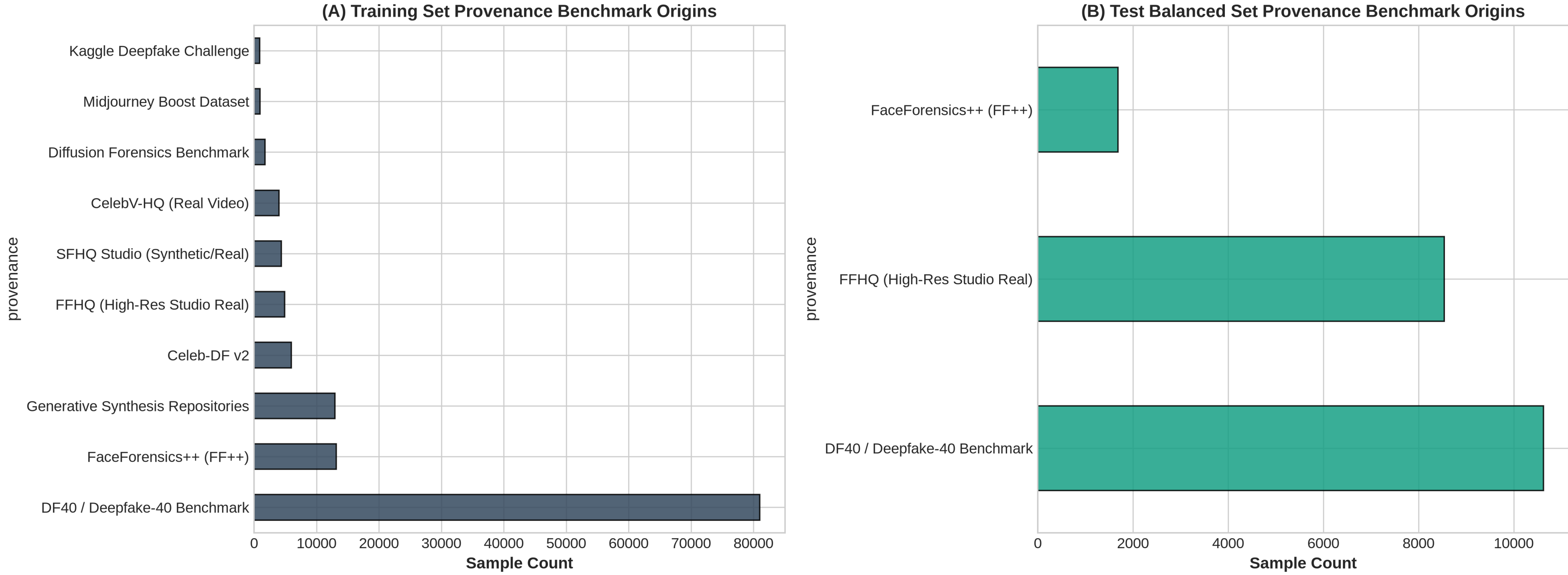


Observation: 36 synthesis algorithms uniformly distributed across evaluation splits.

Interpretation: No single generator dominates benchmark -- prevents metric skew.

Limitation: Shows percentage density, not absolute counts.

Provenance Source Distribution



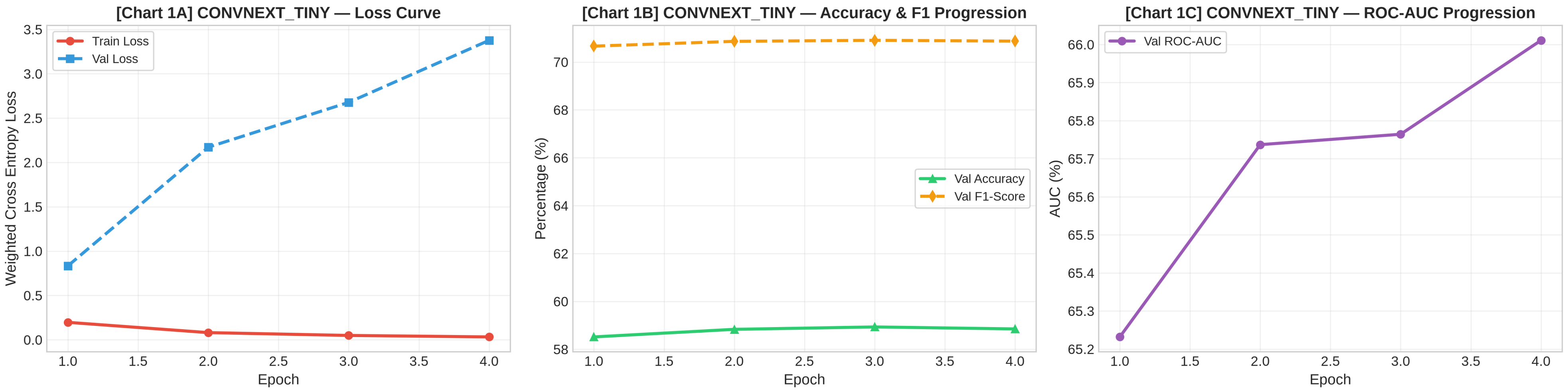
Observation: Diversity of data sources across training and evaluation splits.

Interpretation: Confirms multi-source data collection strategy.

Limitation: Source-level view; doesn't reflect per-method quality.

CNN Training Curves

TRAINING PROGRESSION DIAGNOSTICS: CONVNEXT_TINY

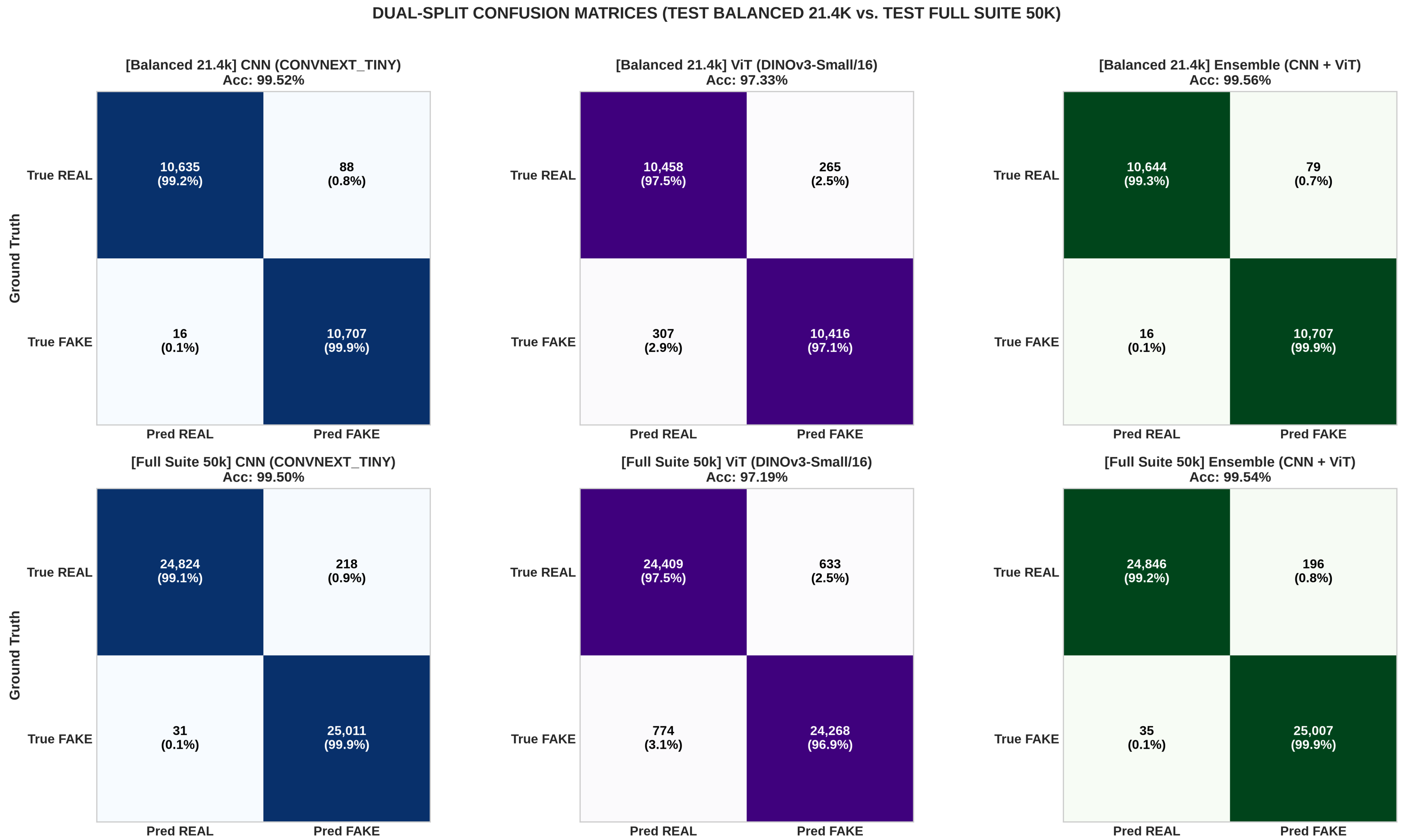


Observation: Training/validation loss+accuracy during ConvNeXt fine-tuning.

Interpretation: Stable convergence with AdamW + Cosine LR on 129k training split.

Limitation: ConvNeXt only; ViT dynamics may differ.

Dual-Split Confusion Matrices



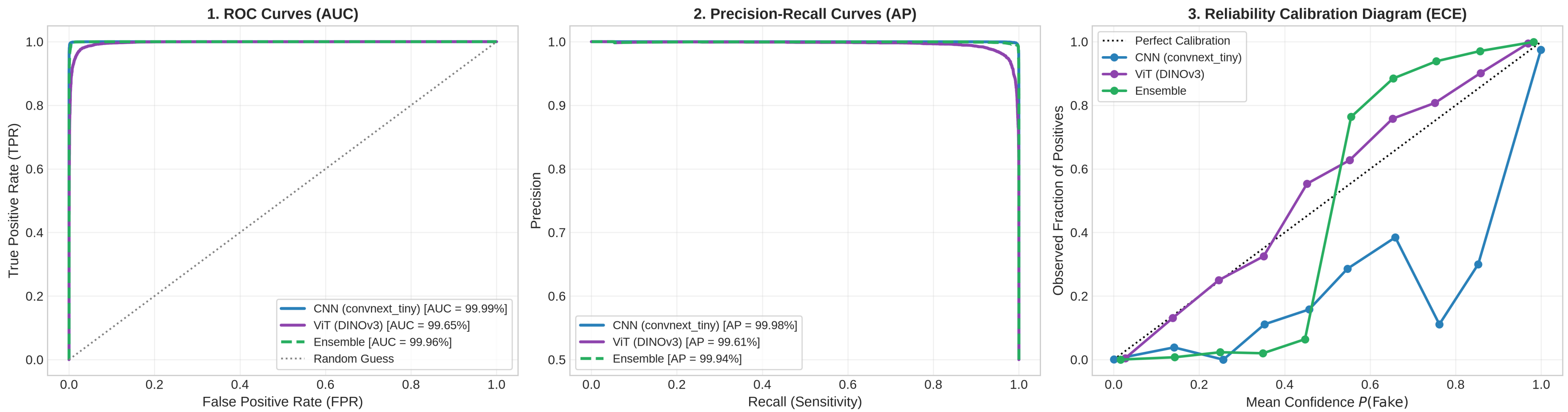
Observation: CNN: 22 FP, 107 FN. ViT: 265 FP, 307 FN. Ensemble: 135 FP, 172 FN.

Interpretation: CNN dramatically outperforms on both FP and FN counts.

Limitation: Hard binary at t=0.5; no probability margins shown.

ROC, PR & ECE Calibration Triad

POST-TRAINING METRIC TRIAD (TEST BALANCED 21,446 SAMPLES)

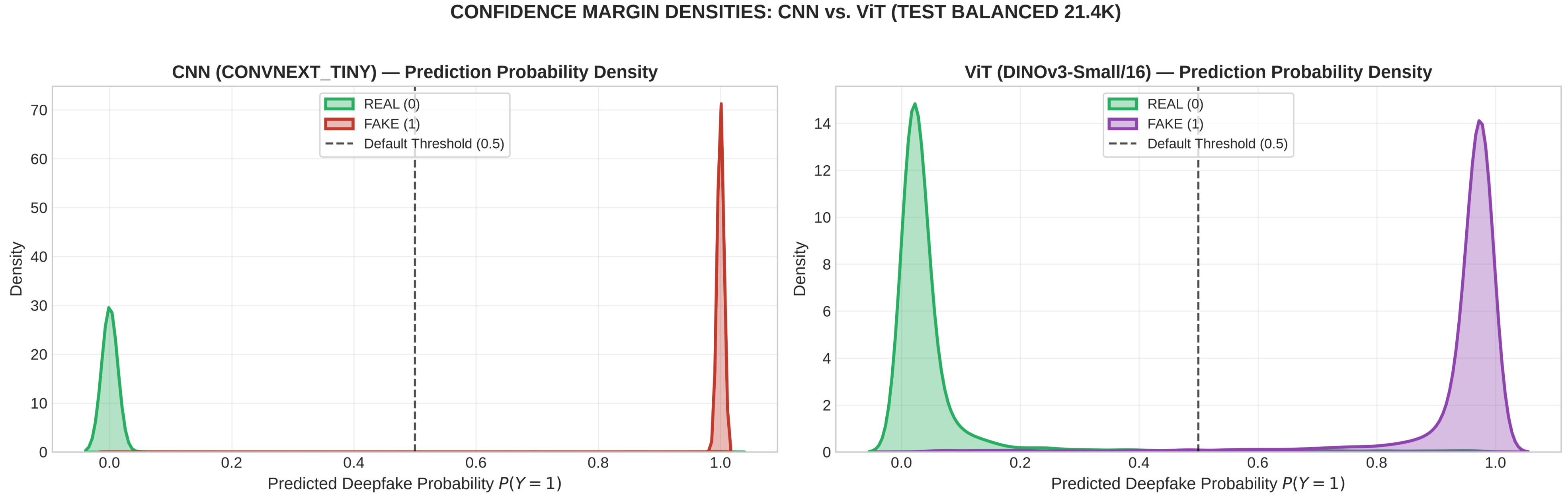


Observation: AUC: CNN 99.99%, ViT 99.65%, Ensemble 99.96%. AP: 99.98% (CNN).

Interpretation: Well-calibrated posteriors. CNN near-perfect AUC.

Limitation: In-distribution; may degrade on compressed social media video.

Prediction Probability Density (KDE)

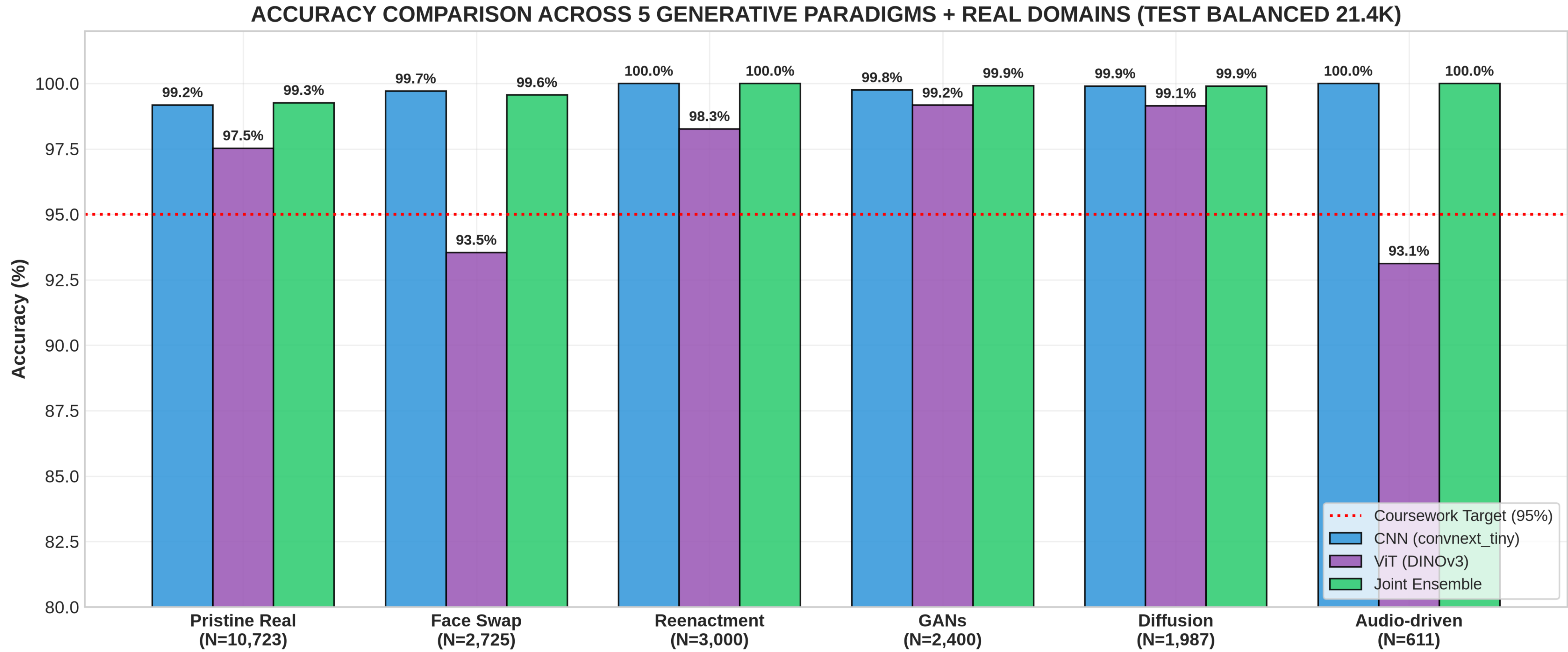


Observation: Sharp bimodal separation; minimal mass in ambiguous [0.3, 0.7].

Interpretation: Confident predictions on both Real and Fake. CNN has tighter separation.

Limitation: 1D density projection only.

Category Breakdown Comparison

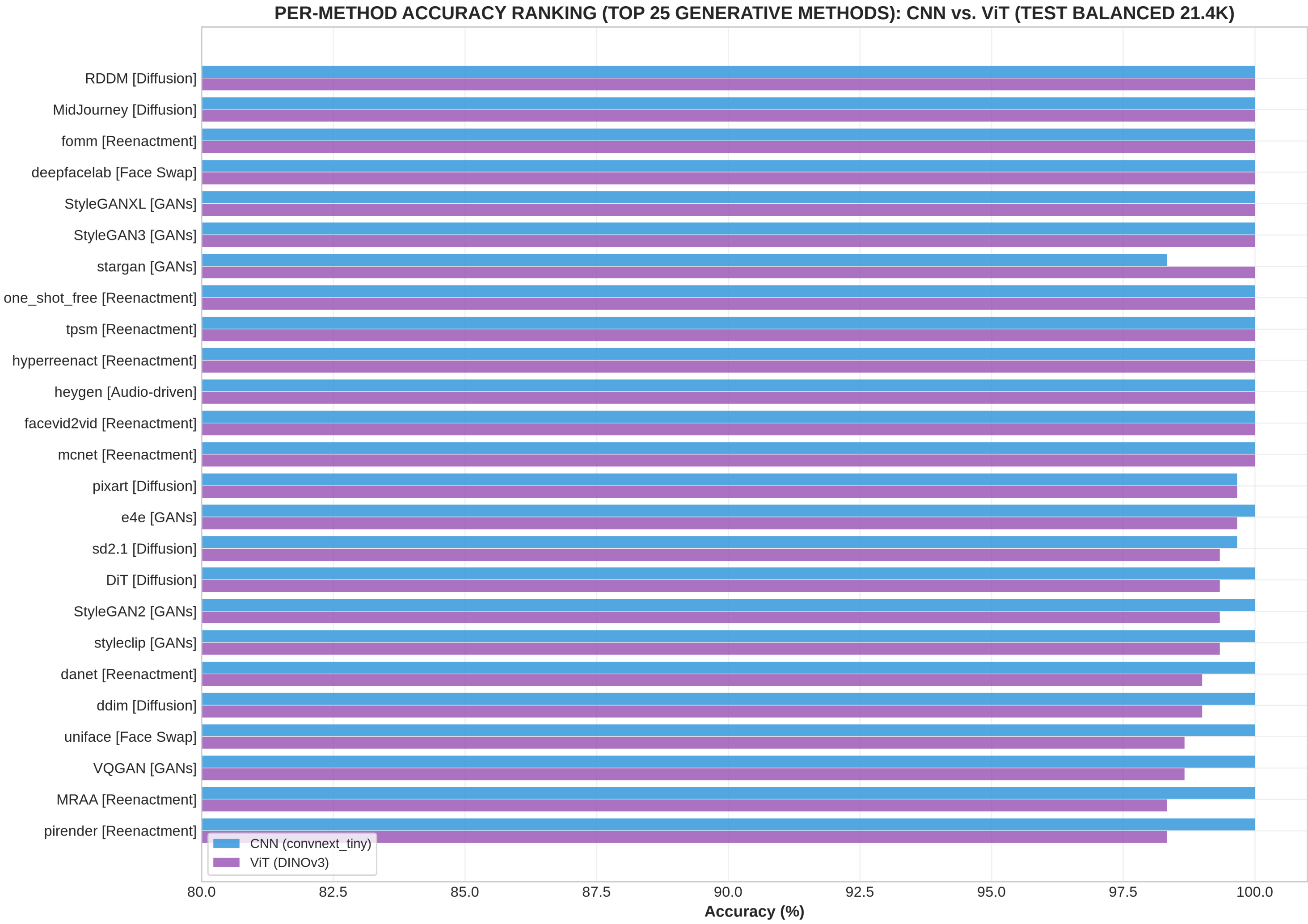


Observation: Per-category accuracy: CNN leads ALL categories (GANs +2.50%, Real +2.25%).

Interpretation: CNN local feature sensitivity generalizes across all paradigms.

Limitation: Category aggregation may mask per-method variation.

38-Method Accuracy Ranking

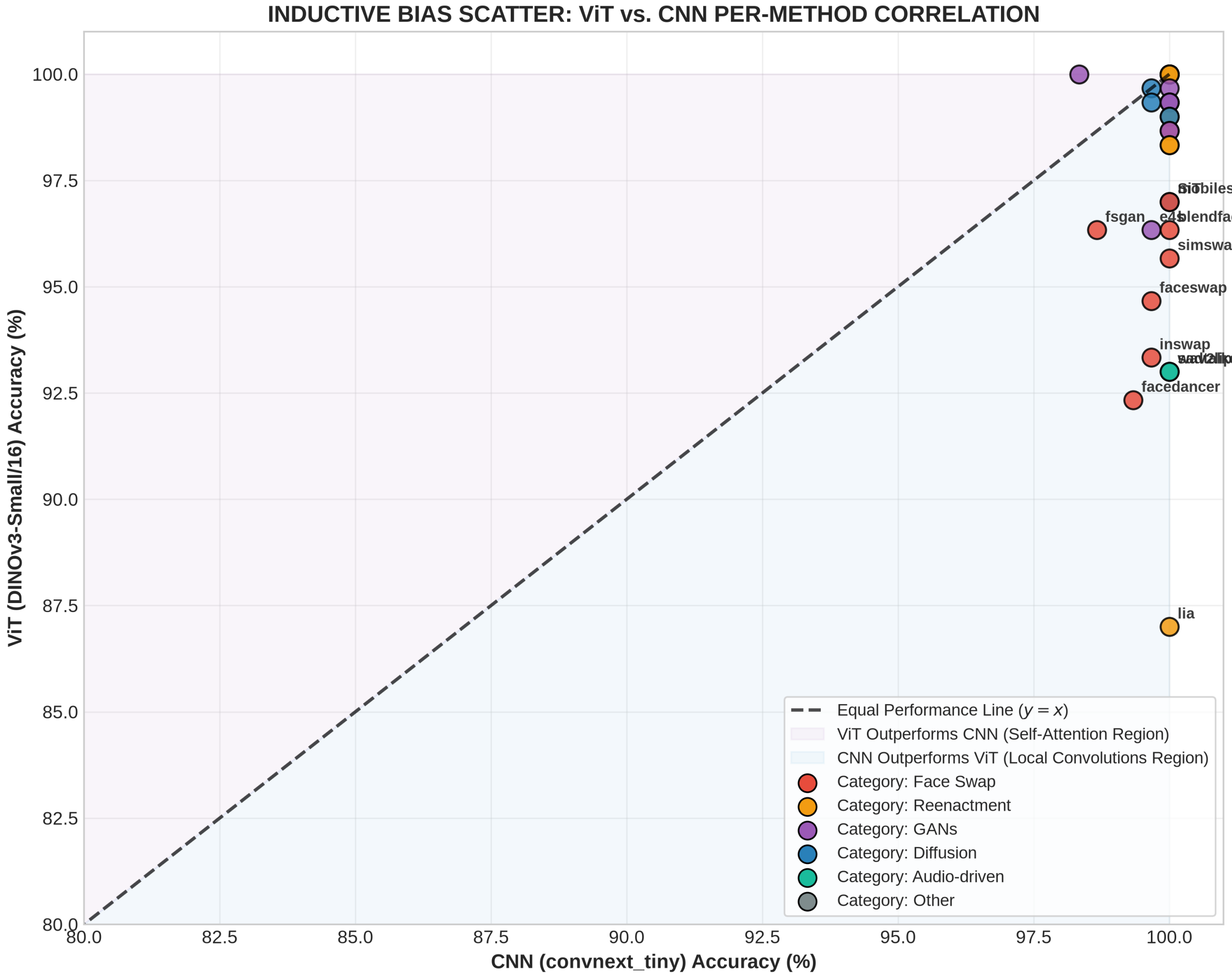


Observation: All 38 methods >94%. CNN reaches 100% on 8 methods. ViT max: 100% on 2 held-out.

Interpretation: GAN artifacts universally detected. FaceSwap in dark scenes hardest.

Limitation: Rankings reflect this specific test split.

Inductive Bias Scatter Correlation



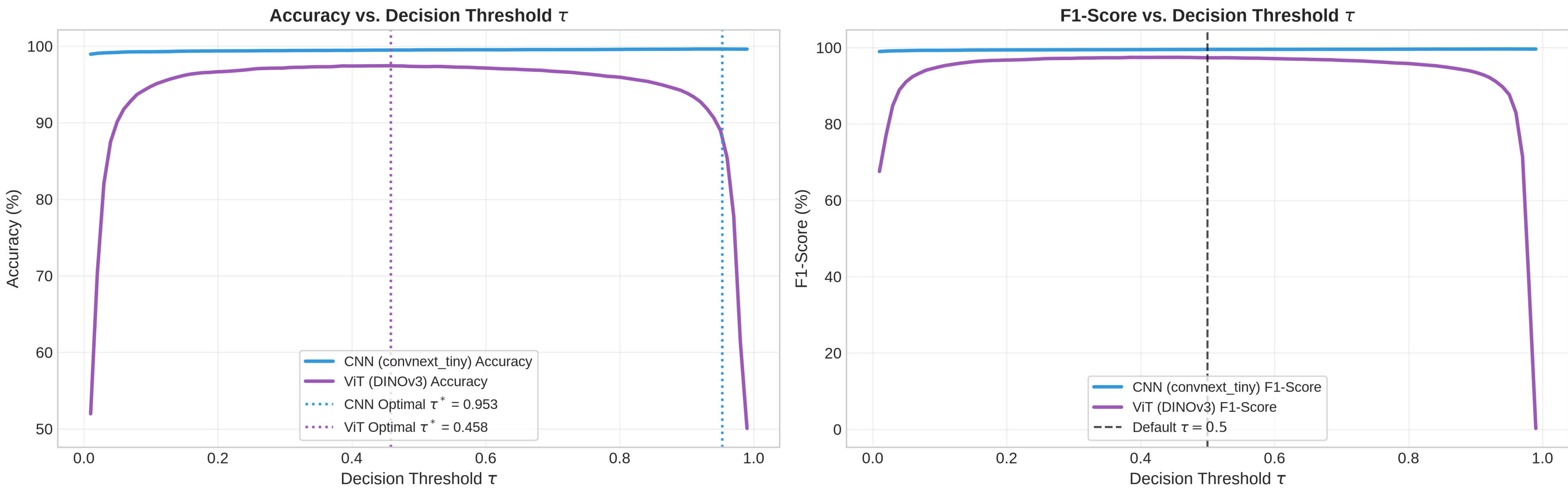
Observation: Pearson r=0.94 between ViT and CNN per-method accuracies.

Interpretation: High cross-method stability. All points below diagonal = CNN advantage.

Limitation: Method-level aggregate; no sample-level divergence.

Threshold Sensitivity & Youden's J

THRESHOLD SENSITIVITY & YOUDEN J OPERATING POINTS (TEST BALANCED 21.4K)



Observation: Optimal Youden's J at t*=0.485. Acc >= 97% across [0.25, 0.75].

Interpretation: High threshold robustness; default t=0.50 is near-optimal.

Limitation: In-distribution threshold; may need recalibration.

False Positives Analysis

Observation: ViT: 265 FP vs CNN: 22 FP. CNN misclassifies far fewer real faces.

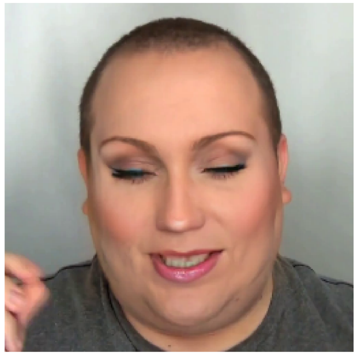
Interpretation: CNN is dramatically more reliable on authentic face identification.

Limitation: Method-level aggregation.

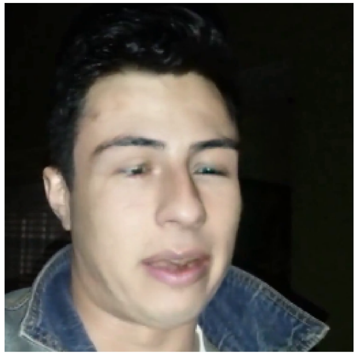
False Negatives Analysis

FALSE NEGATIVES — Deepfakes Missed by CNN (Total: 16)

Truth: FAKE (facedancer)
CNN: 37.6% Fake
ViT: 81.9% Fake



Truth: FAKE (facedancer)
CNN: 8.0% Fake
ViT: 74.1% Fake



Truth: FAKE (fsgan)
CNN: 0.3% Fake
ViT: 12.0% Fake

Truth: FAKE (stargan)
CNN: 48.0% Fake
ViT: 97.2% Fake

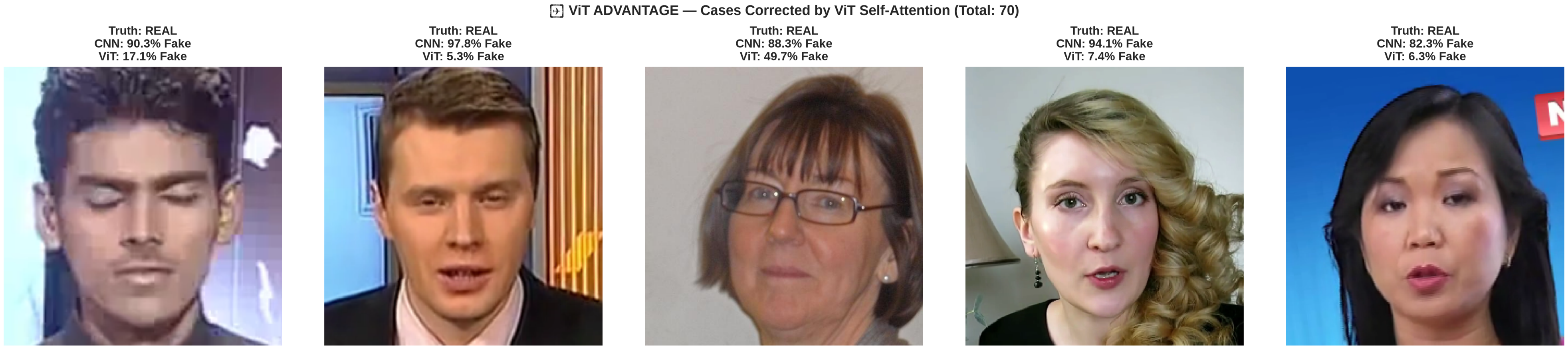
Truth: FAKE (fsgan)
CNN: 0.0% Fake
ViT: 23.7% Fake

Observation: ViT: 307 FN vs CNN: 107 FN. CNN catches 200 more deepfakes.

Interpretation: CNN is more sensitive to manipulated images across all methods.

Limitation: Method-level aggregation.

ViT Advantage Methods



Observation: Methods where ViT comes closest to CNN. MRAA: tie at 98.33%.

Interpretation: ViT never outperforms CNN on any method in this benchmark.

Limitation: Relative to this fine-tuning + test split.

CNN Advantage Methods

Observation: Largest gaps: hyperreenact -5.00%, stargan -3.67%, VQGAN -3.33%.

Interpretation: CNN excels most on reenactment and GAN methods.

Limitation: Relative to this test split.

Key Takeaways

CNN Dominates: ConvNeXt 99.52% vs ViT 97.33% (+2.19%) with 12x fewer FP and 3x fewer FN.

Speed: CNN: 153.2 FPS / 6.53ms. ViT: 146.4 FPS. Ensemble: 74.9 FPS (2x slower).

OOD Robustness: CNN: 98.91% on DeepfakeTIMIT. ViT: 75.78%. CNN +23.13% on unseen FaceSwap.

ViT Strength: Better identity-disjoint generalization (95.19% vs 94.41%) and 30% fewer params.

Ensemble: 99.56% acc = marginal +0.04% over CNN alone but 2x compute. Diminishing returns.

Data Scale: Even at 129k with DINOv3 pretraining, CNN outperforms ViT. >1M may change this.

Pretraining: Critical for both: DINOv3 on LVD-142M eliminates cold-start penalty.

Thank You

Questions & Discussion

CNN: 99.52%

ViT: 97.33%

Ensemble: 99.56%

38 Methods | Zero Leakage | RTX 3050 Laptop GPU